Deepak Mallampati

APPLIED AI ENGINEER

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US-based, F-1 OPT (open to sponsorship) - Remote-US

PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years architecting and shipping production LLM, RAG, and multi-agent systems for regulated enterprises. Build AI-driven product experiences on LangChain/LangGraph - multi-agent orchestration (42% token reduction), RAG over large document corpora (92% precision, 40% fewer hallucinations), and embeddings/vector search with cross-encoder re-ranking. Ship end to end across Python and TypeScript, including a live consumer product with real-time visual/multimodal processing. Balance latency, cost, and quality with rigorous evaluation (RAGAS, BERTScore) at enterprise scale (100k+ requests/day, 99.9% uptime).

CORE COMPETENCIES & TECHNICAL SKILLS

LLMs & Agents (LangGraph / LangChain) • Multi-Agent Systems • RAG Architecture • Embeddings & Vector Search • Multimodal / Vision Pipelines • Python & TypeScript • Model Evaluation & Optimization • Cloud & MLOps

Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain / LangGraph | Multi-Agent Systems | RAG Architecture | Prompt & Tool Orchestration | Fine-tuning (LoRA, QLoRA, PEFT) | Hugging Face Transformers

Retrieval & Multimodal: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Embeddings | Document Chunking | Multimodal AI, Image Analysis, Transformer-based Vision Pipelines

Languages & APIs: Python, JavaScript / TypeScript, Java, SQL | FastAPI, React | REST APIs, Data Pipelines

Machine Learning & NLP: PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | RAGAS Evaluation | BERTScore | Model Optimization | Drift Detection

MLOps & Cloud: AWS (EC2, S3, Lambda, Bedrock, SageMaker), Azure, GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD | MLflow, W&B | Model Monitoring & Observability

Data Engineering: PostgreSQL, MongoDB, Redis | Spark, Databricks | Kafka | Pandas, NumPy

PROFESSIONAL EXPERIENCE

Progress Solutions — *Jul 2025 - Present | USA*

AI Engineer

- Architected production-grade **agentic LLM systems** on a conductor-subagent multi-agent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom metrics) with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Optimized inference cost and latency through prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.
- Built fault-tolerant automation infrastructure with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.

Vanguard
Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration by 40%.
- Optimized LLM prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered content controls for secure, policy-compliant AI responses, reducing unsafe outputs by 42% while maintaining quality.
- Partnered across product, platform, and data engineering to deliver scalable LLM applications supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University
Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 with a curated instruction dataset for controllable long-form generation, conditioning on a six-stage narrative schema.
- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining 99% of baseline predictive performance.

Emerson
Jun 2022 - Apr 2023 | India

ML Trainee

- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific datasets, achieving **89% F1** on custom NER tasks.
- Developed regression and classification models on operational data for equipment-failure prediction, anomaly detection, and capacity forecasting.

PROJECTS

MangaLens
Shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation of manga and webtoons across **29+ sites** - a production **TypeScript** product combining inline visual/multimodal processing with an LLM translation pipeline, turning a multi-step manual chore into a single inline action.

career-ops
Autonomous, AI-driven pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - a multi-agent system built on modern AI-native tooling.

Video Summarization & Highlight Pipeline
End-to-end multimodal pipeline integrating speech-to-text, transformer-based abstractive summarization, and timestamp-aligned clip extraction; deployed across **5,000+ videos**, reducing manual editing effort by 70%.

EDUCATION

M.S. in Computer Science, *Kent State University, OH* — *2025 | GPA 3.70 / 4.0*